Chawakarn Aekanabadin

063-505-3949 | shensmartboyz@gmail.com | github.com/itzshen

Education

Chulalongkorn University — Bangkok, Thailand
B.Eng. Computer Engineering - GPA: 3.72 — *2025 - Present*

Technical Skills

Languages: Java, Python, C, C++, MQL5
Developer Tools: Git, Gradle, JavaFX, DigitalOcean, VS Code, PyCharm, IntelliJ
Libraries: Pandas, NumPy

Projects

NQ Opening Range Breakout Execution EA | *MetaTrader 5, MQL5*

- Developed an automated Opening Range Breakout trading system in MQL5 (inspired by findings from a published research paper) and backtested it on multi-year, 1-minute Nasdaq-100 CFD data in MetaTrader 5's Strategy Tester
- Improved profit factor from baseline 1.15 to 1.96 while reducing trade frequency by approximately 75% through regime filtering and parameter-sensitivity analysis across 2019–2025 data
- Implemented DST-aware session timing and long/short execution with configurable entry cutoffs, stop-loss/ take-profit logic, and fixed-cash or equity-based risk sizing
- Implemented rolling opening-range width percentiles and Kaufman Efficiency Ratio regime filters to evaluate filter effectiveness before enabling trade restrictions
- Designed and ran parameter-sensitivity sweeps, in-/out-of-sample tests, and yearly walk-forward evaluations

myCourseVille Deadline Parser | *Web Scraping, Python, Git*

- Developed a Python CLI tool to fetch assignment deadlines via undocumented API endpoints
- Implemented session-cookie authentication that reuses the browser's existing logged-in session
- Implemented request rate limiting and error handling to reduce unnecessary server requests and handle unexpected API responses
- Formatted deadlines into standard .ics calendar files ready to be imported into Google Calendar

Mining Tycoon Game | *Java, JavaFX, OOP, Gradle, JUnit, Git*

- Designed a modular 2D factory simulation game using object-oriented principles, with interconnected machine and inventory systems
- Implemented a two-step, tick-based grid update system to resolve movement conflicts cleanly and manage item state updates across grid cells
- Developed JUnit tests to validate methods' behavior and handle edge cases

Competitions & Activities

KGI Algo Trading Competition 2026 | *Python, NumPy, Pandas, Git* — (Upcoming) November 2026

- Developed a backtest harness around the bundled tradeSim engine to replay historical market tick data and simulate stock trading across SET50 equities
- Validated backtester accuracy by verifying trade execution, fee calculations, and performance metrics directly against the organizers' official testing engine
- Evaluated seven trading signals on historical SET50 data and excluded signals whose apparent profitability disappeared after transaction costs